

LABORATORY 01

WEB GIS

Your First Web Map

Duration: 2 hours ·

Institute of Geographical Information Systems

School of Civil and Environmental Engineering

National University of Sciences and Technology, Islamabad

Instructor: **Dr. Shahid Nawaz Khan**

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1. Before You Begin

1.1 What You Will Have at the End

A working web page with an interactive map on it, published at a public URL that you can open on your phone and send to anyone.

You will not understand every line of it yet. That is intentional. By week 4 you will be able to write all of it from scratch.

1.2 Learning Outcomes

1. Set up a web development environment on Windows.
2. Explain what HTML, CSS and JavaScript each do.
3. Write a web page and view it through a local web server.
4. Use the browser Developer Tools to inspect what a page loads.
5. Publish a site to the web using Git and GitHub Pages.

1.3 What You Need

Item	Requirement
Operating system	Windows 10 or 11
Software	A web browser (Chrome or Edge), Visual Studio Code, Git
Accounts	A GitHub account
Free disk space	About 500 MB
Administrator rights	Only if you are installing on your own laptop

You were asked to do two things before this lab

Create a GitHub account at github.com/signup, and install Visual Studio Code from code.visualstudio.com.

If you did both, you will finish this lab comfortably. If you did not, do them now in Part 1 and work quickly, because the lab is only two hours.

1.4 Timing

Two hours is tight. Use this to check whether you are on pace.

Time	Part	What you are doing
0:00 to 0:20	1	Set up the tools
0:20 to 0:45	2	Your first web page
0:45 to 1:05	3	Styling with CSS
1:05 to 1:25	4	Add the map
1:25 to 1:50	5	Publish it to the web
1:50 to 2:00	6	Submission and questions

2. Set Up Your Tools

Target: finish by 0:20.

2.1 Visual Studio Code

Skip to 1.2 if it is already installed. On the lab machines it is.

1. Go to `code.visualstudio.com` and download the Windows installer.
2. Run it. Accept the defaults, but tick **Add to PATH** if the option appears.
3. Launch VS Code.

2.2 The Live Server Extension

This is the most important step in Part 1. Do not skip it.

1. In VS Code, click the **Extensions** icon in the left bar. It looks like four squares.
2. Search for **Live Server**.
3. Install the one by Ritwick Dey. It has millions of downloads.

Why you need this, and why opening the file directly will not work

You could open an HTML file by double-clicking it. The address bar would then show `file:///C:/...` and the page would appear to work.

It will break in week 2. When a page loads a data file, the browser blocks that request for pages opened from the file system, as a security measure. Your map data will silently fail to load and the error message will not explain why.

Live Server runs a real web server on your own machine, so the address becomes `http://127.0.0.1:5500`. Everything behaves the way it will on a real server. Get into this habit now.

2.3 Git

Check whether Git is already installed. Open PowerShell by pressing the Windows key, typing `powershell`, and pressing Enter. Then run:

```
git --version
```

If it prints a version number, move on. If it reports that the command is not recognised, install it:

```
winget install --id Git.Git -e
```

Close PowerShell and open a new one, then check the version again.

2.4 Tell Git Who You Are

Run these three commands. Use your own name and email, and use the same email as your GitHub account.

```
git config --global core.autocrlf false
git config --global user.name "Your Name"
git config --global user.email "your.email@student.nust.edu.pk"
```

The first line is not optional

By default, Git on Windows rewrites line endings when it saves files. Later in the semester that silently corrupts scripts that run on Linux servers, producing errors that mention a strange character or claim a file does not exist when you can see it on screen.

That command switches the behaviour off. It costs two seconds now and saves an hour in week 13.

2.5 Create Your Project Folder

In PowerShell:

```
cd C:\
mkdir webgis
cd C:\webgis
code .
```

VS Code opens with C:\webgis as the project folder. Leave it open for the rest of the semester.

Where you put this folder matters

Use C:\webgis exactly. Do not use Documents, Desktop or any folder inside OneDrive.

Documents and Desktop are synced to OneDrive on most Windows laptops. OneDrive locks files while it syncs, which will interfere with your work from week 5 onward in ways that look like random corruption. Spaces in the folder path cause a second, separate set of problems later.

Checkpoint 1

Screenshot your VS Code window showing the empty C:\webgis folder open in the Explorer panel, with the Live Server extension visible as installed.

3. Your First Web Page

Target: finish by 0:45.

3.1 Create the File

In VS Code, click the New File icon in the Explorer panel and name it exactly:

```
index.html
```

The name matters. A web server looks for `index.html` by default when someone visits a folder. Call it anything else and your published site will show a file listing instead of your page.

3.2 Type the Page

Type this out rather than copying it. You will remember the structure far better, and it is only fifteen lines.

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="utf-8">
  <title>My First Web GIS</title>
</head>
<body>

  <h1>Islamabad</h1>
  <p>My first web page for the Web GIS course at IGIS, NUST.</p>

</body>
</html>
```

3.3 View It

Right-click anywhere in the editor and choose **Open with Live Server**. Your browser opens automatically.

Look at the address bar. It should read something like `http://127.0.0.1:5500/index.html`, not `file:///C:/webgis/index.html`. If it says file, you opened the file directly instead of using Live Server. Go back and try again.

Now change the heading text in VS Code and press `Ctrl+S`. The browser reloads by itself. Leave both windows visible side by side for the rest of the lab.

3.4 Understanding What You Typed

Line	What it does
<code><!DOCTYPE html></code>	Tells the browser this is modern HTML. Always the first line
<code><html lang="en"></code>	Wraps the whole document and declares the language
<code><head></code>	Information about the page. Not displayed to the user
<code><meta charset="utf-8"></code>	How to decode the characters. Without it, non-English text breaks
<code><title></code>	The text shown on the browser tab
<code><body></code>	Everything the user actually sees
<code><h1></code>	A top-level heading
<code><p></code>	A paragraph

Notice the pattern. Almost every tag comes in a pair: an opening tag and a closing tag with a

forward slash. Whatever sits between them is what the tag applies to. Tags nest inside each other but must never overlap.

3.5 Open Developer Tools

Press F12 in the browser. This panel is the single most useful tool in this course and we will use it almost every week.

1. Click the **Elements** tab. You are looking at your own HTML, as the browser understood it.
2. Hover over a line and watch the matching part of the page highlight.
3. Click the **Console** tab. It is empty because nothing has gone wrong yet.
4. Click the **Network** tab and reload the page. You will see one request, for `index.html`.

Remember that number: **one request**. Check it again at the end of the lab.

Checkpoint 2

Screenshot your browser showing the page, with Developer Tools open on the Elements tab, and with the address bar visible showing `127.0.0.1`.

4. Styling with CSS

Target: finish by 1:05.

HTML says what exists. CSS says how it looks. You are about to add both a visual style and, more importantly, the empty box that the map will live in.

4.1 Add a Style Block

Add the following inside the head section, immediately after the title line.

inside <head>

```
<style>
  body {
    font-family: system-ui, sans-serif;
    margin: 0;
    padding: 24px;
    background: #f4f6f8;
    color: #13293d;
  }
  h1 {
    margin: 0 0 4px;
    color: #0b2545;
  }
  .subtitle {
    color: #5a6b7b;
    margin-top: 0;
  }
  #map {
    height: 480px;
    width: 100%;
    border-radius: 8px;
  }
</style>
```

4.2 Add the Map Container

Replace the contents of your body section with this:

inside <body>

```
<h1>Islamabad</h1>
<p class="subtitle">Web GIS, Lab 01. IGIS, NUST.</p>

<div id="map"></div>
```

Save and look at the browser. You will see the heading and the subtitle, and nothing else. The div is there, it is 480 pixels tall, and it is empty and invisible. That is correct for now.

4.3 How CSS Finds Things

Three kinds of selector appear above, and this is the whole idea of CSS.

Selector	Written as	Matches
By tag name	body, h1	Every element of that type on the page
By class	.subtitle	Any element with class="subtitle". Reusable, many per page
By id	#map	The one element with id="map". Unique, one per page

The one CSS rule that breaks maps for everybody

A map container with no height is invisible. An empty div has zero height by default, so if you forget the `height` line, Leaflet draws your map into a box that is zero pixels tall and you see nothing at all.

The page will not report an error. The console will be clean. It simply will not appear.

When a map does not show up, check the height of its container first. This will happen to you at least once this semester, and now you know where to look.

4.4 Inspect Your Own CSS

Back in Developer Tools, on the Elements tab, click on the div with id `map`. The Styles panel on the right shows exactly which rules the browser applied to it. Change the height value there directly and watch the page respond. Your file is not modified, so refresh to undo.

Checkpoint 3

Screenshot your page showing the styled heading and subtitle, with the Elements panel open and the map div selected so its CSS rules are visible.

5. Add the Map

Target: finish by 1:25.

Read this before you start

You are about to add code you have not been taught. That is deliberate.

This is a preview of week 3, so that you leave today with a real map rather than a heading and an empty box. Type it, run it, and see it work. You are not expected to understand it yet, and you will not be examined on it this week.

In week 3 we take every line of it apart.

5.1 Load the Leaflet Library

Add this inside the head section, before your style block.

inside <head>

```
<link rel="stylesheet"
      href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css">
<script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
```

Leaflet is about 40 kilobytes of JavaScript written by other people, which knows how to draw maps. Those two lines fetch it from the internet each time your page loads.

5.2 Add the Script

Add this at the end of the body, immediately before the closing body tag.

just before </body>

```
<script>
  const map = L.map('map').setView([33.6844, 73.0479], 12);

  L.tileLayer('https://tile.openstreetmap.org/{z}/{x}/{y}.png', {
    maxZoom: 19,
    attribution: '&copy; OpenStreetMap contributors'
  }).addTo(map);

  L.marker([33.6423, 72.9906])
    .addTo(map)
    .bindPopup('NUST H-12, Islamabad')
    .openPopup();
</script>
```

Save. You should now have a working, pannable, zoomable map of Islamabad with a marker on the NUST campus.

5.3 A Rough Idea of What Those Lines Do

Line	Roughly
L.map('map')	Find the element with id map and turn it into a Leaflet map
.setView([33.68, 73.05], 12)	Centre it on Islamabad at zoom level 12
L.tileLayer(...)	Fetch background map squares from OpenStreetMap
{z}/{x}/{y}	Placeholders. Leaflet fills in zoom, column and row for every tile it needs
L.marker([33.64, 72.99])	Put a pin at the NUST campus
.bindPopup(...)	Attach a label that appears when the pin is clicked

5.4 Now Look at the Network Tab Again

This is the most important thing you will do today. Open Developer Tools, go to the **Network** tab, tick **Disable cache**, and reload the page.

In Part 2 you saw one request. Count them now. You should see dozens.

1. Find the requests ending in `.png`. Click one and open the **Preview** tab.
2. You are looking at a single 256 by 256 pixel square of map.
3. Look at that request's URL. Find the three numbers in it.
4. Now drag the map to pan it, and watch new requests arrive as you drag.

This is exactly what we discussed in the lecture. Your map is not a picture. It is a conversation with a server, and it is happening in front of you.

Checkpoint 4

Screenshot your working map with the Network tab open, showing the tile requests, with one PNG request selected and its Preview tab visible.

5.5 If Your Map Does Not Appear

What you see	Cause	Fix
Nothing at all where the map should be	The map container has no height	Check the <code>#map</code> height rule in your CSS
Grey box, no map imagery	No internet, or the CDN is blocked	Check that you can open <code>unpkg.com</code> in a tab
Console error: <code>L is not defined</code>	The Leaflet script tag is missing, misspelled, or placed after your own script	Leaflet must load before your code runs
Map container not found	The id in the HTML and the id in <code>L.map()</code> do not match	Both must be exactly <code>map</code>
Map appears but the marker is in the sea	Latitude and longitude are swapped	Leaflet wants <code>[latitude, longitude]</code>

Leaflet and GeoJSON disagree about coordinate order

In the lecture we saw that GeoJSON writes `[longitude, latitude]`. Leaflet's own functions take `[latitude, longitude]`, the other way round.

This is a genuine inconsistency in the field, not a mistake in this handout. It catches everybody. When a point lands somewhere absurd, swapping the two numbers is always the first thing to try.

6. Publish It

Target: finish by 1:50.

Your map currently exists only on your own machine. Nobody else can see it. Now you put it on the internet, free, in about fifteen minutes.

6.1 Create the Repository on GitHub

1. Go to `github.com` and sign in.
2. Click the plus icon at the top right, then **New repository**.
3. Name it `webgis-<your-roll-number>`, for example `webgis-2026msrsgis01`.
4. Set it to **Public**. This is required for free GitHub Pages hosting, and it lets me grade it.
5. Do not tick any of the initialise options. Leave README, .gitignore and licence unticked.
6. Click **Create repository**.

Leave the page that appears open. You need the URL from it in a moment.

6.2 Push Your Work

In PowerShell, from your project folder:

```
cd C:\webgis
git init
git add -A
git commit -m "Lab 01: first web map"
git branch -M main
git remote add origin https://github.com/<your-username>/webgis-<roll>.git
git push -u origin main
```

Replace the URL on the `remote` line with your own, copied from the GitHub page you left open.

A browser window will pop up asking you to sign in

This is normal and expected. Git for Windows uses a credential manager that authenticates through your browser rather than asking for a password in the terminal. Approve it once and it remembers.

GitHub stopped accepting account passwords in the terminal some years ago, so if anything asks you to type your GitHub password into PowerShell, something has gone wrong. Ask for help rather than guessing.

6.3 What Those Commands Did

Command	What it did
<code>git init</code>	Turned the folder into a repository that tracks changes
<code>git add -A</code>	Staged every file, meaning marked it to be included in the next save
<code>git commit -m "..."</code>	Saved a permanent snapshot with a message describing it
<code>git branch -M main</code>	Named the main line of work <code>main</code>
<code>git remote add origin</code>	Recorded where the online copy lives
<code>git push</code>	Uploaded your snapshots to GitHub

6.4 Turn On GitHub Pages

1. On your repository page, click **Settings**.
2. In the left sidebar, click **Pages**.
3. Under Source, choose **Deploy from a branch**.
4. Set the branch to `main` and the folder to `/` (root). Click **Save**.
5. Wait. The first deployment takes one to three minutes.

Refresh the Pages settings page until it shows your live URL. It will look like this:

```
https://<your-username>.github.io/webgis-<roll>/
```

6.5 Open It On Your Phone

Open that URL on your phone, on mobile data rather than campus wifi.

Your map is now being served from a real web server to a device on the other side of the room, over the public internet. Two hours ago you had an empty folder.

Checkpoint 5

Screenshot your live GitHub Pages URL open in a browser, with the address bar clearly visible showing the `github.io` address.

7. Submission

7.1 Write Your Report

In VS Code, create a folder called `reports` and inside it a file called `lab01.md`. Put your five checkpoint screenshots in it and answer the questions below.

7.2 Questions

1. In Part 2 your page made one network request. After Part 4 it made dozens. Explain in two or three sentences what changed and why.
2. What is the difference between what HTML does and what CSS does? Give one example of each from your own file.
3. Why does the `#map` rule need a height, when the `h1` rule does not?
4. You opened your page through Live Server at `127.0.0.1` instead of double-clicking the file. Give one reason this matters.
5. A classmate's marker appears in the sea near Africa instead of in Islamabad. What is almost certainly wrong, and how would you fix it?

7.3 Push and Submit

```
cd C:\webgis
git add -A
git commit -m "Lab 01: report"
git push
```

Then paste two things into the LMS submission box: your repository URL, and your live GitHub Pages URL.

7.4 Marking

Item	Marks
Working page with a visible map	30
Five checkpoint screenshots	25
Live GitHub Pages URL that loads	20
Answers to the five questions	25

This repository is now your semester project

Every lab from here adds a commit to this same repository. By week 16 it will hold a full Web GIS application with a database behind it.

Your commit history is part of your grade. Fourteen weekly commits tell a very different story from one commit made the night before the deadline, and the dates are visible to me. Commit as you work.

8. If You Finish Early

These are optional and unmarked. They are the fastest way to get ahead of week 3.

8.1 Try These

1. Move the marker to your own home town. Find the coordinates by right-clicking on Google Maps and copying what appears, then remember to check the order before you paste it in.
2. Add a second marker with a different popup message.
3. Change the starting zoom from 12 to 16, then to 8. Watch how many tile requests each one makes.
4. Swap the tile layer for a different basemap. Search for a Stamen or CartoDB tile URL and substitute it.
5. Add `L.circle([33.6423, 72.9906], {radius: 2000}).addTo(map);` and see what happens.

8.2 Command Reference

<code>cd C:\webgis</code>	go to your project folder
<code>code .</code>	open the folder in VS Code
<code>git status</code>	what has changed since the last commit
<code>git add -A</code>	stage all changes
<code>git commit -m "..."</code>	save a snapshot with a message
<code>git push</code>	upload to GitHub
<code>git log --oneline</code>	see your commit history

8.3 Rules to Remember

- Always open pages through Live Server, never by double-clicking the file.
- The project folder is `C:\webgis`. Never OneDrive, Documents or Desktop.
- When a map does not appear, check the container height first.
- Leaflet uses `[latitude, longitude]`. GeoJSON uses `[longitude, latitude]`.
- Commit and push before you leave the lab, every week.

8.4 Before Next Week

Nothing to install. Bring the same laptop. We move on to GeoJSON, coordinate systems, and why file size decides whether your map is usable.